

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **Toluene-d8**  
 Product Description: **Toluene-d8**  
 Cat No. : 166390000; 166390100; 166390250  
 Synonyms: Methylbenzene-d8  
 CAS No: 2037-26-5  
 Molecular Formula: C7 D8

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use: Laboratory chemicals.  
 Uses advised against: No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**E-mail address**

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**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                    |
|------------------------------------------------------|--------------------|
| Flammable liquids                                    | Category 2 (H225)  |
| Aspiration Toxicity                                  | Category 1 (H304)  |
| Skin Corrosion/Irritation                            | Category 2 (H315)  |
| Reproductive Toxicity                                | Category 2 (H361d) |
| Specific target organ toxicity - (single exposure)   | Category 3 (H336)  |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373)  |
| Chronic aquatic toxicity                             | Category 3 (H412)  |

**Label Elements**


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## Signal Word

## Danger

### Hazard Statements

H225 - Highly flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H315 - Causes skin irritation  
H336 - May cause drowsiness or dizziness  
H361d - Suspected of damaging the unborn child  
H373 - May cause damage to organs through prolonged or repeated exposure if inhaled  
H412 - Harmful to aquatic life with long lasting effects

### Precautionary Statements

#### Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection  
P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray

#### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P331 - Do NOT induce vomiting  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component    | CAS No    | Weight % |
|--------------|-----------|----------|
| (2H8)Toluene | 2037-26-5 | >95      |
| Toluene      | 108-88-3  | -        |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

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|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

## **Most important symptoms and effects, both acute and delayed**

. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Causes central nervous system depression.

## **Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## **SECTION 5: FIREFIGHTING MEASURES**

### **Extinguishing media**

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

### **Special hazards arising from the substance or mixture**

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### **Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

Use personal protective equipment as required. Ensure adequate ventilation.

### **Environmental precautions**

Should not be released into the environment.

### **Methods and Material for Containment and Cleaning Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

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## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Ensure adequate ventilation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame. Store under an inert atmosphere. Keep container tightly closed in a dry and well-ventilated place. Protect from moisture.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | Malaysia | ACGIH TLV   | OSHA PEL                                                                                                                                                               |
|-----------|----------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Toluene   |          | TWA: 20 ppm | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 375 mg/m <sup>3</sup><br>Ceiling: 300 ppm<br>(Vacated) STEL: 150 ppm<br>(Vacated) STEL: 560 mg/m <sup>3</sup><br>TWA: 200 ppm |

| Component | European Union                                                                                                                | The United Kingdom                                                                                                        | Germany                                                                                                                                                                                                                                                                    |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Toluene   | TWA: 50 ppm (8hr)<br>TWA: 192 mg/m <sup>3</sup> (8hr)<br>STEL: 100 ppm (15min)<br>STEL: 384 mg/m <sup>3</sup> (15min)<br>Skin | STEL: 100 ppm 15 min<br>STEL: 384 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 191 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 50 ppm (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 190 mg/m <sup>3</sup> (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK<br>TWA: 190 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 100 ppm<br>Höhepunkt: 380 mg/m <sup>3</sup><br>Haut |

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

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Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

## Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Appearance     | Colorless                |
| Physical State | Liquid                   |
| Odor           | aromatic                 |
| Odor Threshold | No data available        |
| pH             | No information available |

|                     |                    |
|---------------------|--------------------|
| Melting Point/Range | -84 °C / -119.2 °F |
| Softening Point     | No data available  |
| Boiling Point/Range | 110 °C / 230 °F    |
| Flash Point         | 4 °C / 39.2 °F     |

Method - No information available

|                          |                                              |        |
|--------------------------|----------------------------------------------|--------|
| Evaporation Rate         | No data available                            |        |
| Flammability (solid,gas) | Not applicable                               | Liquid |
| Explosion Limits         | <b>Lower</b> 1.2 Vol%<br><b>Upper</b> 7 Vol% |        |

|                              |                                     |             |
|------------------------------|-------------------------------------|-------------|
| Vapor Pressure               | 29.1 hPa @ 20°C                     |             |
| Vapor Density                | No data available                   | (Air = 1.0) |
| Specific Gravity / Density   | 0.940                               |             |
| Bulk Density                 | Not applicable                      | Liquid      |
| Water Solubility             | 0.5 g/L @20°C practically insoluble |             |
| Solubility in other solvents | No information available            |             |

### Partition Coefficient (n-octanol/water)

|           |                |
|-----------|----------------|
| Component | <b>log Pow</b> |
| Toluene   | 2.73           |

|                           |                          |                                             |
|---------------------------|--------------------------|---------------------------------------------|
| Autoignition Temperature  | 535 °C / 995 °F          |                                             |
| Decomposition Temperature | No data available        |                                             |
| Viscosity                 | No data available        |                                             |
| Explosive Properties      |                          | Vapors may form explosive mixtures with air |
| Oxidizing Properties      | No information available |                                             |

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Molecular Formula C7 D8  
Molecular Weight 100.21

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Hygroscopic.

### Possibility of Hazardous Reactions

**Hazardous Polymerization** Hazardous polymerization does not occur.  
**Hazardous Reactions** None under normal processing.

### Conditions to Avoid

Incompatible products. Excess heat. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

Strong oxidizing agents. Strong acids. Halogens. Halogenated compounds.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

**Product Information** No acute toxicity information is available for this product

**(a) acute toxicity;**  
**Oral** No data available  
**Dermal** No data available  
**Inhalation** No data available

| Component | LD50 Oral            | LD50 Dermal            | LC50 Inhalation       |
|-----------|----------------------|------------------------|-----------------------|
| Toluene   | > 5000 mg/kg ( Rat ) | 12000 mg/kg ( Rabbit ) | 26700 ppm ( Rat ) 1 h |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**  
**Respiratory** No data available

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|                                                                                                         |                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin</b>                                                                                             | No data available                                                                                                                                                                                              |
| <b>(e) germ cell mutagenicity;</b>                                                                      | No data available                                                                                                                                                                                              |
| <b>(f) carcinogenicity;</b>                                                                             | No data available<br>There are no known carcinogenic chemicals in this product                                                                                                                                 |
| <b>(g) reproductive toxicity;<br/>Reproductive Effects<br/>Developmental Effects<br/>Teratogenicity</b> | No data available<br>Experiments have shown reproductive toxicity effects on laboratory animals.<br>Developmental effects have occurred in experimental animals.<br>Possible risk of harm to the unborn child. |
| <b>(h) STOT-single exposure;</b>                                                                        | No data available                                                                                                                                                                                              |
| <b>Results / Target organs</b>                                                                          | Central nervous system (CNS).                                                                                                                                                                                  |
| <b>(i) STOT-repeated exposure;</b>                                                                      | No data available                                                                                                                                                                                              |
| <b>Target Organs</b>                                                                                    | Neuropsychological effects, Eyes, Ears.                                                                                                                                                                        |
| <b>(j) aspiration hazard;</b>                                                                           | Category 1                                                                                                                                                                                                     |
| <b>Other Adverse Effects</b>                                                                            | The toxicological properties have not been fully investigated.                                                                                                                                                 |
| <b>Symptoms / effects, both acute and delayed</b>                                                       | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Causes central nervous system depression.                                                 |
| <b>Endocrine Disrupting Properties</b>                                                                  | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.                                                                            |

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains a substance which is: Toxic to aquatic organisms.

| Component | Freshwater Fish                                                                                              | Water Flea                                                                                         | Freshwater Algae                                                                                                                 | Microtox                |
|-----------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Toluene   | 50-70 mg/L LC50 96 h<br>5-7 mg/L LC50 96 h<br>15-19 mg/L LC50 96 h<br>28 mg/L LC50 96 h<br>12 mg/L LC50 96 h | EC50: = 11.5 mg/L, 48h<br>(Daphnia magna)<br>EC50: 5.46 - 9.83 mg/L,<br>48h Static (Daphnia magna) | EC50: = 12.5 mg/L, 72h static<br>(Pseudokirchneriella subcapitata)<br>EC50: > 433 mg/L, 96h<br>(Pseudokirchneriella subcapitata) | EC50 = 19.7 mg/L 30 min |

**Persistence and degradability** Readily biodegradable  
**Persistence** Persistence is unlikely.

| Component               | Degradability |
|-------------------------|---------------|
| Toluene<br>108-88-3 (-) | 86% (20d)     |

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Toluene   | 2.73    | 90                            |

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**Mobility in soil** The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Spillage unlikely to penetrate soil. The product is insoluble and floats on water. Is not likely mobile in the environment due its low water solubility.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used Do not flush to sewer Can be landfilled or incinerated, when in compliance with local regulations

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**  
UN-No UN1294  
Hazard Class 3  
Packing Group II  
Proper Shipping Name TOLUENE

**Road and Rail Transport**  
UN-No UN1294  
Hazard Class 3  
Packing Group II  
Proper Shipping Name TOLUENE

**IATA**  
UN-No UN1294  
Hazard Class 3  
Packing Group II  
Proper Shipping Name TOLUENE

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Inventories** X = listed



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| Component    | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------|-----------|------|-----|-------|------|------|-------|------|----------|
| (2H8)Toluene | 218-009-5 | -    | -   | -     | -    |      | X     | X    | -        |
| Toluene      | 203-625-9 | X    | X   | X     | X    | X    | X     | X    | KE-33936 |

| Component | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Toluene   |                                                                                           |                                                                                          |                            | Annex I - Y42                      |

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information

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relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**